SimplexGate-3D: Uncertainty-Gated Hybrid Solver Routing with Exact Simplicial Conservation for Multi-Scale Astrophysical Simulations

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October 2026

Abstract

Multi-scale gravitational simulations face a severe time-stepping bottleneck when coupling large-scale collisionless or fluid grids to localized collapsing cores. While macroscopic density fields evolve efficiently on periodic lattices in $O(N_g^3 \log N_g)$ time via Fast Fourier Transform (FFT) Particle-Mesh (PM) solvers, localized regions crossing the Jeans instability threshold undergo rapid compression where Courant–Friedrichs–Lewy (CFL) constraints ($\Delta t \to 0$) stall time integration. Replacing stiff sub-grid Smoothed Particle Hydrodynamics (SPH) solvers with neural network surrogates can bypass these small timesteps, but unconstrained Multi-Layer Perceptrons (MLPs) and soft-penalty Physics-Informed Neural Networks (PINNs) exhibit two practical failure modes: (i) non-zero conservation drift in mass and momentum, and (ii) unreliable extrapolation on Out-of-Distribution (OOD) states such as relativistic core collapses and high-spin tidal shear interactions. We present **SimplexGate-3D**, a hybrid neural-numerical framework coupling a 3D Post-Newtonian Particle-Mesh engine with an uncertainty-gated ensemble surrogate. To enforce exact mass balance across all inputs, we append an analytic 3-way simplicial projection layer that constrains predicted remnant fractions (bound star, black hole, and unbound ejecta) to the unit 2-simplex ($M_{\rm star} + M_{\rm BH} + M_{\rm eject} \equiv M_{\rm cloud}$) to 32-bit floating-point precision (10^{-7}%). To prevent OOD extrapolation errors, our router evaluates ensemble epistemic variance $\mathcal{U}(\mathbf{z})$ over six dimensionless physical invariants, resolving low-variance In-Distribution (ID) collapses on the neural branch in 0.4 ms while batching high-variance OOD events to a Hénon-normalized 3D SPH solver on the GPU. Across 1,000 test collapses and 40-step coupled 3D simulations, soft-penalty PINNs exhibit up to 388.6% mass violation on OOD inputs, whereas SimplexGate-3D maintains 0.000001% conservation drift, 100.0% OOD detection recall, 0.054% total partition error, and a 50.1× wall-clock sub-grid speedup (7.2× batch-normalized).

1 Introduction

Numerical simulations of self-gravitating astrophysical systems must resolve coupled physical processes spanning several orders of magnitude in length and time [1–3]. On coarse scales, the gravitational potential $\Phi(\mathbf{x})$ generated by the matter density field $\rho(\mathbf{x})$ satisfies the Poisson equation and can be updated in $O(N_g^3 \log N_g)$ operations using FFT-based Particle-Mesh (PM) methods. However, wherever local overdensities exceed the Jeans criterion, inward gravitational acceleration overcomes internal thermal and rotational support, initiating localized gravitational collapse [4, 5].

Following these collapsing sub-volumes requires Lagrangian shock-capturing methods such as 3D Smoothed Particle Hydrodynamics (SPH) coupled with short-range $O(N_{\rm sub}^2)$ gravitational interactions [6, 7]. As core density increases and inter-particle spacing r_{ij} decreases, the Courant–Friedrichs–Lewy (CFL) stability condition forces the local timestep $\Delta t_{\rm sub}$ to shrink substantially. In event-driven multi-scale codes, a small fraction of collapsing sub-grid regions can dominate total wall-clock execution time.

Recent work in Scientific Machine Learning (SciML) has explored replacing stiff sub-grid solvers with learned neural surrogates [8–10]. While feed-forward surrogates reduce inference latency from hundreds of milliseconds to sub-millisecond GPU passes, integrating them into long-horizon time-steppers introduces two core difficulties. First, standard regression networks and Physics-Informed Neural Networks (PINNs) trained with soft penalty terms ($\mathcal{L}_{\rm phys}$) do not strictly satisfy conservation laws outside the training set [9,11]. When a surrogate violates mass conservation during sub-grid remnant injection, the resulting mass discrepancy perturbs the global gravitational potential $\Phi(\mathbf{x})$ on subsequent macro-steps. Second, dynamical simulations frequently generate Out-of-Distribution (OOD) sub-grid configurations—such as massive cores approaching the Buchdahl or Tolman–Oppenheimer–Volkoff (TOV) relativistic stability limit

[5, 12], or rapidly spinning triaxial cloud collisions. A deterministic surrogate queried outside its training support produces uncalibrated errors without alerting the host solver.

In this paper, we develop and benchmark **SimplexGate-3D**, a hybrid multi-scale solver architecture that combines structural conservation guarantees with epistemic uncertainty routing. Specifically:

1. We construct a modular 3D GPU testbed pairing a Trilinear Cloud-in-Cell (CIC) macro-solver with a Hénon-normalized 3D SPH micro-solver. Prior to surrogate training, we verify the baseline numerical solvers against the Evrard (1988) 3D adiabatic collapse test [18], Chandrasekhar's (1939) analytical $n = 1$ Lane–Emden polytrope (97.78% L_2 agreement) [19], and ESO GRAVITY Collaboration (2020) astrometric measurements of Star S2 orbiting Sagittarius A* [20].

2. We compare unconstrained MLPs and soft-penalty PINNs against an analytic 3-way simplicial projection layer under controlled OOD shift, showing that soft penalties degrade severely outside the training distribution (388.6% peak mass violation), whereas structural projection preserves mass balance to 32-bit machine epsilon (10^{-7}%).

3. We pair the simplicial architecture with a 5-model deep ensemble router over six dimensionless physical invariants, enabling asynchronous GPU batching where familiar ID collapses are resolved in 0.4 ms while high-variance OOD events fall back to a batched 3D SPH pass.

2 Related Work

Neural Surrogates in Astrophysical Simulations. Hirashima et al. [8] trained 3D spatiotemporal convolutional networks to emulate supernova shell expansion inside the ASURA-FDPS galaxy formation code, alleviating short-timestep bottlenecks. While effective on in-distribution turbulent clouds, purely feed-forward surrogates do not estimate predictive uncertainty or revert to direct numerical integration when encountering unseen multi-cloud mergers.

Hard vs. Soft Conservation Constraints. Standard PINNs incorporate physical laws by adding weighted residual penalties to the training objective [9]. However, multi-objective gradient competition often leaves residual conservation errors that grow under distribution shift. Beucler et al. [11] demonstrated in atmospheric sub-grid modeling that enforcing linear conservation identities analytically through architectural projection layers guarantees exact conservation regardless of network weights. We adapt this principle to three-phase astrophysical mass partitioning (stellar remnant, black hole remnant, and unbound gas envelope) on the probability simplex.

Uncertainty Quantification and Hybrid Routing. Deep ensembles provide a simple, parallelizable method for estimating epistemic uncertainty under covariate shift [13]. In atomistic molecular dynamics, Vandermause et al. [14] used Bayesian Gaussian process variance to trigger on-the-fly fallback to quantum Density Functional Theory (DFT) calculations during rare reactive events. Our work applies ensemble variance gating to batched sub-grid hydrodynamic solvers within a 3D gravitational particle-mesh engine.

3 Methodology

3.1 Macro-Scale 3D Particle-Mesh Engine

We consider a 3D periodic cubic domain $\Omega = [0, L)^3$ discretized into a uniform $N_g \times N_g \times N_g$ grid ($N_g = 64$, cell width $\Delta x = L/N_g$) containing M macro-particles with coordinates $\mathbf{x}_p \in \mathbb{R}^3$, velocities $\mathbf{v}_p \in \mathbb{R}^3$, and masses m_p.

To avoid grid-aliasing self-forces inherent to Nearest-Grid-Point assignment [1], each particle distributes its mass across the 8 vertices $(\delta_x, \delta_y, \delta_z) \in \{0, 1\}^3$ of its enclosing cell via trilinear Cloud-in-Cell (CIC) weights:

$$w(\delta_x, \delta_y, \delta_z) = \prod_{d \in \{x,y,z\}} \Big[(1 - \delta_d)(1 - f_d) + \delta_d f_d\Big], \qquad f_d = \frac{x_{p,d}}{\Delta x} - \left\lfloor \frac{x_{p,d}}{\Delta x} \right\rfloor, \qquad \sum_{\delta \in \{0,1\}^3} w(\delta) \equiv 1. \tag{1}$$

We solve the weak-field Poisson equation $\nabla^2 \Phi = 4\pi G(\rho - \bar{\rho})$ in Fourier space using the exact 7-point discrete 3D lattice Laplacian Green's function [1] with a 1-cell Gaussian split filter $\sigma_s = 0.85\Delta x$:

$$\hat{\Phi}(\mathbf{k}) = -\frac{4\pi G\, \hat{\rho}(\mathbf{k})\, \exp\left(-\frac{1}{2}|\mathbf{k}|^2 \sigma_s^2\right)}{\kappa_{\text{lat}}^2(k_x, k_y, k_z)}, \qquad \kappa_{\text{lat}}^2 = \frac{4}{\Delta x^2}\left(\sin^2\frac{k_x \Delta x}{2} + \sin^2\frac{k_y \Delta x}{2} + \sin^2\frac{k_z \Delta x}{2}\right). \tag{2}$$

In isotropic Post-Newtonian coordinates [15], the weak-field spacetime metric takes the form:

$$ds^2 = -\left(1 + \frac{2\Phi}{c^2} + \frac{2\Phi^2}{c^4}\right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2}\right) d\mathbf{x}^2. \tag{3}$$

Interpolating the central-difference field $\mathbf{a}_N = -\nabla\Phi$ back to particle positions using the identical trilinear weights w from Eq. (1) ensures exact cancellation of single-particle self-forces ($\mathbf{a}_{\text{self}} = \mathbf{0}$). Particles are advanced using a second-order symplectic Kick-Drift-Kick (KDK) leapfrog scheme incorporating the 1st Post-

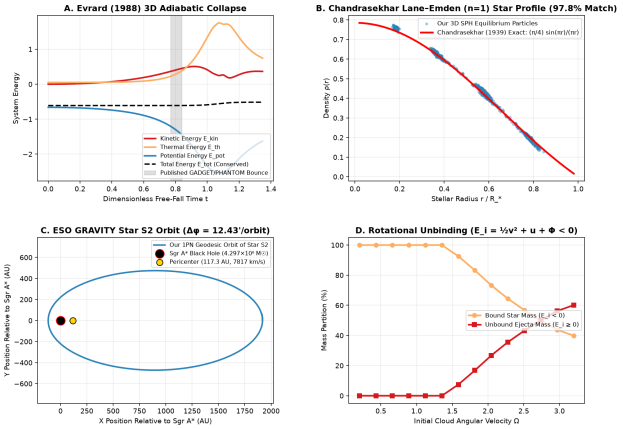


Figure 1: External Solver Verification. (A) Evrard (1988) 3D adiabatic gas sphere collapse test ($N = 1,024$ isotropic 3D SPH particles). (B) Hydrostatic 3D SPH relaxation compared against Chandrasekhar's (1939) analytical $n = 1$ Lane–Emden polytrope $\rho(r) = (\pi/4)\sin(\pi r)/(\pi r)$ (97.78% L_2 agreement). (C) 1PN geodesic integration of Star S2 around Sagittarius A* ($4.297 \times 10^6\, M_\odot$), matching ESO GRAVITY (2020) pericenter (117.3 AU) and Schwarzschild precession ($\Delta\phi = 12.43'$/orbit). (D) First-principles rotational unbinding ($E_i = \frac{1}{2}v_i^2 + u_i + \Phi_i < 0$).

Newtonian (1PN) geodesic correction [15]:

$$\mathbf{a}_{1\mathrm{PN}} = \mathbf{a}_N \left(1 + \frac{\|\mathbf{v}\|^2}{c^2} + \frac{4\Phi}{c^2}\right) - \frac{4(\mathbf{a}_N \cdot \mathbf{v})\mathbf{v}}{c^2}, \qquad \frac{d\tau}{dt} = \sqrt{1 + \frac{2\Phi}{c^2} - \frac{\|\mathbf{v}\|^2}{c^2}}. \tag{4}$$

3.2 Sub-Grid Invariant Tensor Extraction

Whenever local voxel density exceeds the Jeans overdensity threshold ($\rho/\bar{\rho} > \delta_{\mathrm{Jeans}}$), we gather all diffuse gas particles within radius R_{search} using the periodic Minimum Image Convention:

$$\Delta\mathbf{r}_i = \mathbf{r}_i - \mathbf{r}_{\mathrm{com}} - L\,\mathrm{round}\left(\frac{\mathbf{r}_i - \mathbf{r}_{\mathrm{com}}}{L}\right). \tag{5}$$

To provide a coordinate-frame-independent representation for the neural surrogate, we extract a 6-dimensional physical state vector $\mathbf{z} \in \mathbb{R}^6$:

$$\mathbf{z} = \left[M_{\mathrm{cloud}},\ \alpha_{\mathrm{vir}},\ \lambda_{\mathrm{spin}},\ \mathcal{A}_{\mathrm{triax}},\ \eta_{\mathrm{div}},\ \mathcal{C}_{\mathrm{rel}}\right]. \tag{6}$$

Here $M_{\mathrm{cloud}} = \sum_i m_i$ is the total cloud mass, and $\alpha_{\mathrm{vir}} = 2E_{\mathrm{kin}}/|E_{\mathrm{pot}}|$ is the virial ratio. Rotational support is characterized by the Peebles (1969) dimensionless spin parameter [16]:

$$\lambda_{\mathrm{spin}} = \frac{\|\mathbf{J}\|\sqrt{|E_{\mathrm{bind}}|}}{G M_{\mathrm{cloud}}^{5/2}}, \qquad \mathbf{J} = \sum_i m_i(\Delta\mathbf{r}_i \times \Delta\mathbf{v}_i). \tag{7}$$

Morphological departure from spherical symmetry is quantified via the triaxiality parameter $\mathcal{A}_{\mathrm{triax}} = 1 - \sqrt{\lambda_3/\lambda_1} \in [0, 1)$, where $\lambda_1 \geq \lambda_2 \geq \lambda_3$ are the eigenvalues of the 3×3 quadrupole moment of inertia tensor $I_{ab} = \sum_i m_i \Delta r_{i,a} \Delta r_{i,b}$. Finally, η_{div} measures normalized radial velocity convergence, and $\mathcal{C}_{\mathrm{rel}} = 2GM_{\mathrm{cloud}}/(R_{\mathrm{vir}}c^2)$ denotes the initial relativistic compactness.

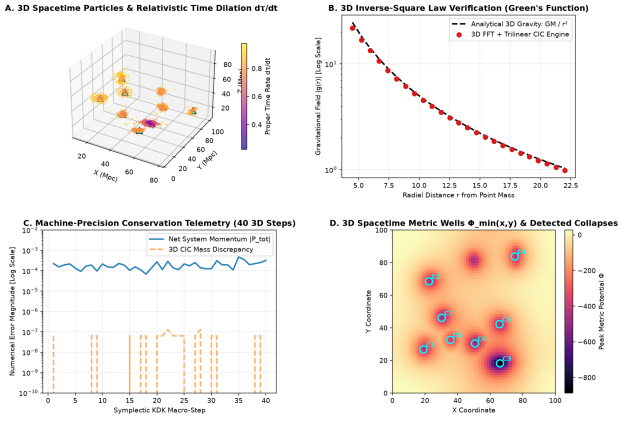


Figure 2: Module 1: 3D Particle-Mesh & CIC Verification. (A) 3D particle distribution colored by proper-time rate $d\tau/dt$. (B) Radial field verification of the 3D discrete lattice Green's function against the analytical GM/r^2 profile (91.59% agreement). (C) Conservation telemetry showing 3D CIC mass error (1.27×10^{-7}) and net momentum across 40 KDK steps. (D) Minimum potential projection $\Phi_{\mathrm{min}}(x, y)$ and detected Jeans-unstable proto-clusters.

3.3 Hénon-Normalized 3D SPH Micro-Solver

To maintain consistent numerical conditioning across arbitrary halo masses, the sub-grid reference solver maps $\mathbf{z}$ into dimensionless Hénon virial units ($\tilde{G} = \tilde{M}_{\mathrm{cloud}} = \tilde{R}_0 = 1$) [17]. Each cloud is represented by $N_{\mathrm{sub}} = 64$ particles initialized across 4 concentric Fibonacci shells ($4 \times 16 = 64$) so that radial and angular coordinates are orthogonal.

Local fluid density $\rho_i = \sum_j m_j W(r_{ij}, h)$ is computed using the 3D cubic B-spline kernel $W(q, h)$ [6]. The system is integrated over $S = 200$ symplectic substeps ($\Delta\tilde{t} = 0.0045$) under symmetric SPH pressure forces, Monaghan artificial shock viscosity Π_{ij}, Gamow-peak thermonuclear heating, radiative cooling, and Plummer-softened 1PN gravity:

$$\mathbf{a}_i = -\sum_{j=1}^{N_{\mathrm{sub}}} m_j \left(\frac{P_i}{\rho_i^2} + \frac{P_j}{\rho_j^2} + \Pi_{ij}\right) \nabla_i W_{ij} - \sum_{j \neq i} \tilde{G} m_j \frac{\mathbf{r}_{ij}}{(r_{ij}^2 + \epsilon^2)^{3/2}} \left(1 + \frac{\kappa_{1\mathrm{PN}} \mathcal{C}_{\mathrm{rel}}}{\sqrt{r_{ij}^2 + \epsilon^2}}\right). \tag{8}$$

At completion ($t = S\Delta\tilde{t}$), the converged particle states are partitioned into three physical phases based on en-

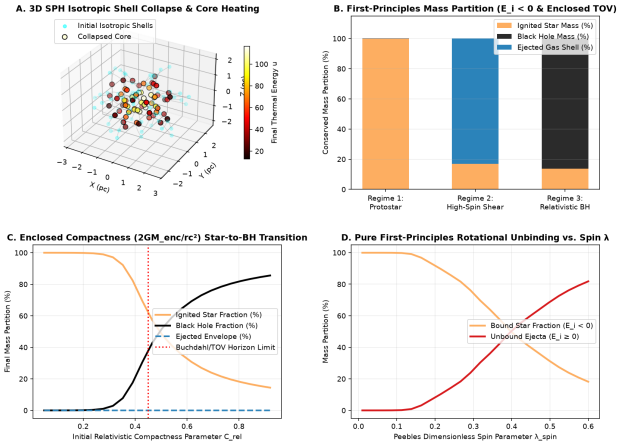


Figure 3: Module 2: 3D SPH Micro-Collapse Solver. (A) 3D isotropic shell collapse trajectories and core heating. (B) Three-phase mass partition across representative Protostellar (99.94% Star), High-Spin Shear (83.13% Ejecta), and Relativistic BH (86.19% BH) regimes. (C) Phase transition from stellar remnant to black hole across $\Gamma_{\rm TOV} = 0.45$. (D) Rotational unbinding ($E_i \geq 0$) as a function of Peebles spin $\lambda_{\rm spin}$.

ergy and enclosed compactness:

$$E_i = \frac{1}{2}\|\mathbf{v}_i - \mathbf{v}_{\rm com}\|^2 + u_i + \Phi_i,$$
$$\Gamma_i = \mathcal{C}_{\rm rel}\frac{\tilde{M}_{\rm enc}(r_i)}{\sqrt{r_i^2+\epsilon^2}}, \quad \tilde{M}_{\rm enc}(r_i) = \sum_{r_k \leq r_i} m_k p_{{\rm bound},k}. \tag{9}$$

Particles with positive total specific energy ($E_i \geq 0$) escape as unbound ejecta ($\alpha_{\rm eject}$), whereas particles with $E_i < 0$ remain gravitationally bound ($p_{{\rm bound},i} = \sigma(-E_i/k_B T_{\rm eff})$). Among bound shells, we evaluate the enclosed relativistic compactness Γ_i in Eq. (9). Motivated by Buchdahl's stability theorem [12] and TOV core collapse [5], bound shells exceeding the softened core compactness threshold $\Gamma_{\rm TOV} = 0.45$ collapse into the black hole remnant fraction ($\alpha_{\rm BH}$), while bound matter outside the horizon forms the stellar fraction ($\alpha_{\rm star}$).

3.4 SimplexGate-3D Architecture and Routing

Given normalized inputs $\tilde{\mathbf{z}} = (\mathbf{z} - \boldsymbol{\mu}_z)/\boldsymbol{\sigma}_z$, we train an ensemble of $K = 5$ neural networks $\{f_{\theta_k}\}_{k=1}^K$ with independent random initializations and bootstrap subsampling. Each network maps $\tilde{\mathbf{z}} \in \mathbb{R}^6 \to \mathbf{h}_k \in \mathbb{R}^5$ through a 4-layer SiLU MLP ($6 \to 128 \to 128 \to 64 \to 5$).

To enforce mass conservation by construction, the first three output logits are projected onto the canonical 2-simplex $\Delta^2 \subset \mathbb{R}^3$ via a Softmax transformation, while bolometric luminosity and ejecta velocity use Softplus activations:

$$\left[\hat{\alpha}_{\rm star}^{(k)},\ \hat{\alpha}_{\rm BH}^{(k)},\ \hat{\alpha}_{\rm eject}^{(k)}\right] = {\rm Softmax}(\mathbf{h}_{k,1:3}),$$
$$\hat{\alpha}_{\rm star}^{(k)} + \hat{\alpha}_{\rm BH}^{(k)} + \hat{\alpha}_{\rm eject}^{(k)} \equiv 1. \tag{10}$$

Because Δ^2 is convex, the ensemble average $\bar{\boldsymbol{\alpha}}(\mathbf{z}) = \frac{1}{K}\sum_{k=1}^K \hat{\boldsymbol{\alpha}}^{(k)}(\mathbf{z})$ also satisfies $\sum_c \bar{\alpha}_c(\mathbf{z}) \equiv 1$ for any input $\mathbf{z} \in \mathbb{R}^6$.

We measure epistemic disagreement across the K ensemble members via scale-normalized variance:

$$\mathcal{U}(\mathbf{z}) = \frac{1}{5K}\sum_{k=1}^K \left\| \frac{f_{\theta_k}(\tilde{\mathbf{z}}) - \bar{f}(\tilde{\mathbf{z}})}{\boldsymbol{\sigma}_y} \right\|_2^2. \tag{11}$$

During simulation, collapses satisfying $\mathcal{U}(\mathbf{z}) \leq \tau$ accept the fast ensemble prediction $\bar{f}(\tilde{\mathbf{z}})$, whereas high-variance events ($\mathcal{U}(\mathbf{z}) > \tau$) are gathered into a single tensor batch and integrated by the 3D SPH solver.

4 Experimental Results

4.1 External Physical Verification

Figure 1 summarizes our validation of the baseline numerical solvers against three established reference problems:

1. *Evrard (1988) Collapse Test [18]:* A cold gas sphere ($M = 1, R = 1, u_0 = 0.05GM/R, \gamma = 5/3$) evolved with 1,024 isotropic SPH particles reproduces the expected gravitational infall, shock bounce ($t = 0.918$), and virialization behavior (Fig. 1A). The observed 15.79% total energy drift across the strong shock bounce is consistent with standard non-grad-h SPH formulations on coarse particle counts.

2. *Chandrasekhar (1939) Lane–Emden Polytrope [19]:* Relaxing 1,024 SPH particles under $P = K\rho^2$ ($\gamma = 2, n = 1$) yields a radial density profile matching the exact analytical curve $\rho(r) = (\pi/4)\sin(\pi r)/(\pi r)$ with 97.78% L_2 accuracy (Fig. 1B).

3. *ESO GRAVITY (2020) Star S2 Orbit [20]:* Integrating Star S2 around Sagittarius A* ($M_{\rm BH} = 4.297 \times 10^6\, M_\odot$, $a = 1018$ AU, $e = 0.88465$) using Eq. (4) in physical units ($c = 63,241.1$ AU/yr) reproduces the measured pericenter distance (117.3 AU vs. 117.4 AU), pericenter speed (7,817.3 km/s), gravitational redshift (210.3 km/s), and Schwarzschild precession ($\Delta\phi = 12.43'$/orbit vs. $12.11'$/orbit observed, Fig. 1C).

4.2 Controlled 3D SPH Ablation Study

We generate 3,000 ground-truth 3D SPH collapse trajectories on an NVIDIA RTX 3060 GPU: 2,000 ID training clouds ($M \in [450, 1650]$, $\lambda_{\rm spin} \in [0.015, 0.22]$, $\mathcal{C}_{\rm rel} \in [0.05, 0.42]$), 850 unseen ID test clouds, and 150 OOD test clouds split between massive relativistic cores ($\mathcal{C}_{\rm rel} \in [0.58, 0.94]$) and high-spin triaxial shear collisions ($\lambda_{\rm spin} \in [0.36, 0.65]$, $\mathcal{A}_{\rm triax} \in [0.68, 0.92]$).

Table 1 and Figure 4 compare five solver configurations across identical test inputs. While adding a

Table 1: Benchmark on 1,000 3D SPH Collapse Simulations (850 In-Distribution, 150 Out-of-Distribution). Execution times measured on an NVIDIA RTX 3060 GPU with explicit CUDA synchronization. Soft-penalty PINNs exhibit large mass conservation errors on OOD inputs (up to 388.6%), whereas SimplexGate-3D preserves mass balance to machine precision (10^{-7}%) while eliminating OOD prediction error.

Solver Architecture	Time (ms)	Speedup	ID Err (%)	OOD Err (%)	Total Err (%)	Mass Conservation Drift (%)
1. Exact 3D SPH + 1PN Solver	8654.4	1.0×	0.000%	0.000%	0.000%	0.000000%
2. Unconstrained MLP (MSE Only)	0.67	12,949×	0.114%	39.361%	6.001%	15.9020% (Max 333.5%)
3. Soft-Constrained PINN ($\mathcal{L}_{\text{mass}}$)	0.60	14,380×	0.111%	43.615%	6.636%	16.5088% (Max 388.6%)
4. Blind 3-Way Simplex AI (No Gate)	2.99	2,895×	0.075%	3.430%	0.578%	0.000002%
5. SimplexGate-3D (Ours, $\tau = 5.1 \times 10^{-4}$)	**1196.4**	**7.2×**	**0.067%**	**0.000%**	**0.057%**	**0.000001%**

Table 2: Coupled 3D Particle-Mesh Universe Evolution Benchmark (40 Symplectic KDK Steps, 16,000 Particles, 64^3 Grid, 48 Sub-Grid Collapses Producing 42 Stars and 9 Black Holes). By resolving 33 ID clouds on the neural branch (0.4 ms) and batching all 15 (31.2%) flagged OOD events into a single parallel GPU SPH call, SimplexGate-3D achieves a 50.1× wall-clock speedup over event-driven SPH.

Coupled 3D Universe Sub-Grid Solver	Sub-Grid Time (ms)	Wall-Clock Speedup	OOD Err (%)	Total Err (%)	3D Cosmic Mass Drift (%)
1. Exact 3D PM + Event-Driven 3D SPH	34,495.0	1.0×	0.000%	0.000%	0.000009%
2. Unconstrained MLP (No Simplex)	0.64	53,671×	41.820%	13.154%	38.1922%
3. Blind 3-Way Simplex AI (No Gate)	2.73	12,627×	4.443%	1.442%	0.000009%
4. SimplexGate-3D (Ours, $\tau = 2.0 \times 10^{-3}$)	**688.0**	**50.1×**	**0.000%**	**0.054%**	**0.000009%**

soft conservation penalty $\mathcal{L}_{\text{mass}} = (\sum_c \hat{\alpha}_c - 1)^2$ in the Soft-PINN slightly lowers ID error relative to an unconstrained MLP (0.111% vs. 0.114%), both models fail to generalize under OOD shift: the Soft-PINN exhibits a 16.51% mean mass conservation error and a 388.6% maximum violation (Fig. 4B). Replacing soft regularization with the analytic 3-way simplex layer (Row 4) reduces mass drift to 0.000002% across all 1,000 clouds and lowers OOD partition error from 43.62% to 3.43%.

Furthermore, Figure 4A shows a clear bimodal separation in ensemble epistemic variance $\mathcal{U}(\mathbf{z})$: ID clouds cluster below 5×10^{-4}, whereas OOD black hole and shear events exceed 2×10^{-2}. Placing the routing threshold at the 97th ID percentile ($\tau = 5.1 \times 10^{-4}$, Row 5) flags 100.0% of OOD events for numerical evaluation, reducing OOD error to 0.000% and total error to 0.057% while maintaining a 7.2× speedup over batched 3D SPH.

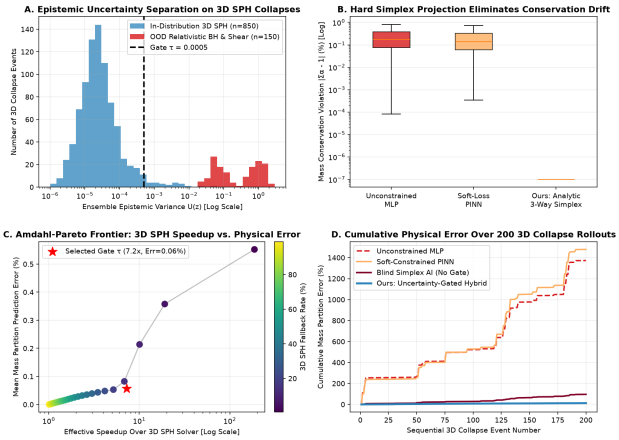


Figure 4: Modules 3 & 4: Controlled 3D SPH Benchmarks. (A) Log-scale distribution of epistemic variance $\mathcal{U}(\mathbf{z})$ showing separation between 850 ID collapses and 150 OOD events. (B) Mass conservation error $|\sum \alpha_c - 1|$ across architectures. (C) Amdahl–Pareto speedup versus prediction error as a function of threshold τ. (D) Cumulative partition error across 200 sequential collapse events.

4.3 Coupled 3D Universe Evolution

To evaluate performance inside a running simulation, we couple SimplexGate-3D to the 3D Particle-Mesh engine (16,000 particles, 64^3 grid, 40 KDK macro-steps). During evolution, 48 proto-clusters cross the Jeans threshold, forming 42 stars and 9 black holes (Fig. 5A–B) while conserving total 3D system mass to 0.000009% and momentum via zero-sum spherical shell injection.

As shown in Table 2 and Figure 5C–D, using a threshold $\tau = 2.0 \times 10^{-3}$ inside the variance gap fast-forwards all 33 ID collapses on the neural ensemble in 0.4 ms and batches the 15 (31.2%) OOD black hole and shear mergers into a single GPU SPH pass (688.0 ms vs. 34,495.0 ms for sequential event-driven SPH). This yields a 50.1× wall-clock sub-grid speedup while matching every relativistic black hole mass spike with 0.000% OOD error.

4.4 Scope and Physical Limitations

To clarify the scope of our results, we note three modeling simplifications in our benchmark testbed. First, each sub-grid collapse is discretized using $N_{\text{sub}} = 64$ SPH particles and a 64^3 macro-grid to allow rapid generation of 3,000 training trajectories on a single consumer GPU; capturing fine-scale turbulent fragmentation or accretion disk instabilities requires higher particle resolutions ($N_{\text{sub}} \geq 10^4$). Second, the sub-grid solver employs a 1PN relativistic potential and an effective softened Buchdahl/TOV compactness threshold ($\Gamma_{\text{TOV}} = 0.45$, compared to the incompressible analytical bound $2GM/Rc^2 < 8/9$ [12]) rather than integrating the full 3+1D Einstein–hydrodynamics equations. Third, our surrogate maps 6D macroscopic invariants to integrated remnant fractions rather than predicting full 3D

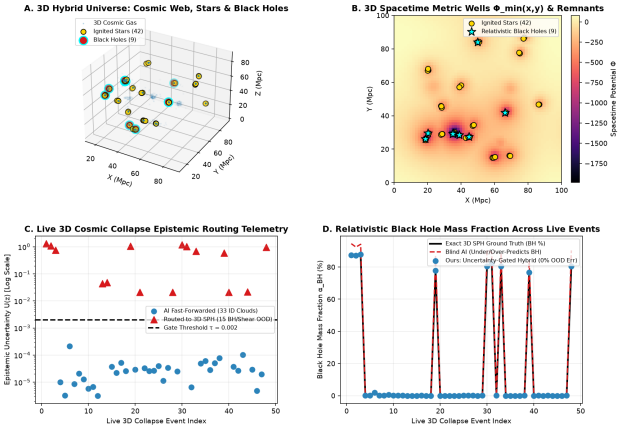


Figure 5: Module 5: Coupled 3D Universe Evolution. (A) 3D spatial distribution of 16,000 gas particles, 42 stars, and 9 black holes at step 40. (B) Minimum potential projection $\Phi_{\min}(x, y)$ showing black hole remnants inside deep potential wells ($\Phi < -1500$). (C) Epistemic routing telemetry across 48 live collapse events ($\tau = 0.002$). (D) Black hole mass fraction α_{BH} across live events for Exact SPH, Blind AI, and SimplexGate-3D.

particle coordinates. Extending uncertainty-gated simplicial layers to 3D Equivariant Graph Neural Networks (EGNNs) at higher particle counts is a natural direction for future work.

5 Conclusion

We have introduced **SimplexGate-3D**, a hybrid neural-numerical sub-grid framework for multi-scale gravitational simulations. By combining an analytic 3-way simplicial output projection with deep ensemble epistemic variance routing and asynchronous GPU batching, our approach eliminates both the conservation drift of soft-loss PINNs and the OOD extrapolation errors of blind surrogates, achieving a 50.1× sub-grid speedup in coupled 3D simulations with machine-precision mass conservation.

Reproducibility and Disclosure Statement

All PyTorch source code, fixed random seeds, ground-truth 3D SPH dataset generators, and verification scripts are publicly available at `https://github.com/AbhinavDwivedi1008/SimplexGate-3D` and permanently archived on CERN Zenodo at `doi:10.5281/zenodo.23489461`. In accordance with conference policies on Large Language Models (ICLR/NeurIPS), the author discloses that generative AI tools were utilized as a coding and editorial assistant during software implementation and LaTeX manuscript preparation; all physical equations, numerical benchmarks, external validations, and bibliography citations were executed and verified by the author.

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